Broadband 2 GHz to 14 GHz Dual-Polarized Horn Antenna With Stable Unidirectional Radiation Patterns

Xianbo Cao, *Member, IEEE*, Pengfei Zhang, Tao Hong, *Senior Member, IEEE*, and Wen Jiang, *Senior Member, IEEE*

***Abstract*—A broadband dual-polarized quad-ridged horn (QRH) antenna covering 2 GHz to 14 GHz is proposed for antenna measurements and electromagnetic compatibility (EMC) testing. To achieve dual-port broadband impedance matching, a metallic ring-shaped matching block and two metallic bridges are incorporated into the feeding cavity. Additionally, the ridge profile is optimized using a fourth-order Bezier curve to ensure smooth impedance transitions, while the sidewall structure is shortened to mitigate radiation pattern distortion at high frequencies. As a result of these enhancements, the fabricated antenna prototype exhibits a reflection coefficient below −10 dB, port isolation exceeding 30 dB, a gain ranging from 6.3 dBi to 19.3 dBi, and stable, symmetrical radiation patterns without beam splitting across the entire operating band. Compared with conventional designs, the proposed antenna achieves both broader bandwidth and improved radiation stability, thereby addressing typical pattern distortion in broadband applications.**

***Index Terms*—Antenna measurement, broadband, dual-polarization, horn antenna, quad-ridged.**

I. Introduction

BROADBAND horn antennas are widely employed in a range of applications, including antenna measureme-nts (serving as both reference and source antennas), electromagnetic compatibility and interference (EMC/I) testing, communication systems, satellite tracking systems, reflector feed systems, and electronic warfare (EW) [1], [2], [3], [4], [5]. Their popularity is largely due to their advantageous characteristics such as high gain, broad bandwidth, strong power-handling capability, stable radiation patterns, ease of excitation, and relatively straightforward construction [6].

The original horn antenna exhibited a bandwidth of 3:1 [7], which was subsequently extended to 12:1 [8], [9]; however, significant radiation pattern distortion was observed. Recent studies have focused on extending the bandwidth of horn antennas while enhancing radiation pattern stability. In [10] and [11], the Bezier curve was employed to model the ridge structure, thereby enhancing aperture matching. In [12], the sidewalls of the horn antenna were replaced with adjustable metallic grids, thereby mitigating radiation pattern discontinuities at high frequencies. A quad-ridged horn (QRH) antenna was designed for radio telescope applications, incorporating a recessed contour profile to achieve greater phase-center stability in the radiation pattern [13]. A cross-shaped inverted-ridge horn antenna was proposed, wherein the conventional metallic back cavity was replaced with a polytetrafluoroethylene (PTFE) cavity to suppress higher-order modes [14]. However, the aforementioned antennas exhibit several limitations, including single polarization, pattern splitting, and relatively high sidelobe levels. According to previous studies [15], [16], broadband horn antennas are prone to higher-order mode excitation, which can result in pattern distortion. Therefore, it remains necessary to develop broadband horn antennas with stable radiation patterns.

In this letter, a dual-polarized broadband QRH antenna with stable radiation characteristics is proposed and analyzed. A ring-shaped matching block is introduced at the base of the feeding cavity to realize broadband performance. Two metallic bridges are introduced below the ridges to enable simultaneous impedance matching at both ports. Furthermore, the ridge profiles are optimized using Bezier curves to ensure smooth impedance transitions. The sidewall structure is also carefully shortened to suppress higher-order mode excitation, thereby further preventing degradation of the radiation pattern. As a result of these enhancements in the feeding cavity and flared section, the proposed horn antenna demonstrates the following performance characteristics:

1) broadband performance with a 7:1 bandwidth ratio;
2) a stable radiation pattern without splitting, exhibiting improved consistency across the entire operating band;
3) reasonable gain, cross-polarization level, and isolation.

II. Antenna Structure Design

This letter employs a modular design methodology, in which the antenna is divided into a feeding cavity and a flared section. Each module is independently designed and analyzed to reduce computational complexity and to accelerate the design process.

A. Feeding Cavity

The structure of the feeding cavity is shown in Fig. 1. A metallic ring-shaped matching block is employed instead of the

Received 5 September 2025; revised 14 October 2025; accepted 9 November 2025. Date of publication 13 November 2025; date of current version 5 February 2026. This work was supported in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation under Grant Number GZC20241300; in part by the Fellowship of China Postdoctoral Science Foundation under Grant Number 2024M762534; and in part by the National Natural Science Foundation of China under Grant 62401444, Grant 62171335, and Grant U23B2017. *(Corresponding author: Xianbo Cao.)*

The authors are witht the Hangzhou Institute of Technology, Xidian University, Hangzhou 311200, China, and also with the National Key Laboratory of Radar Detection and Sensing, Xidian University, Xi'an 710071, China (e-mail: caoxianbo@xidian.edu.cn).

Digital Object Identifier 10.1109/LAWP.2025.3632783

1548-5757 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on September 29,2026 at 10:06:19 UTC from IEEE Xplore. Restrictions apply.

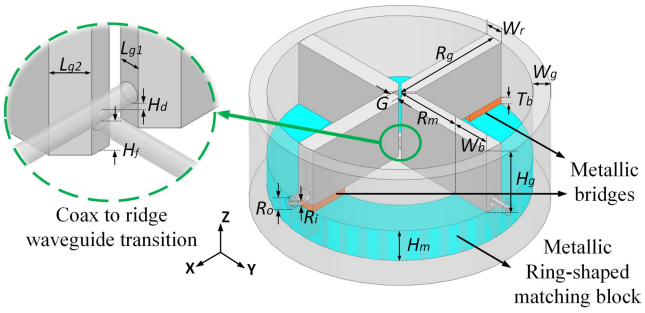


Fig. 1. Structure of feeding cavity.

TABLE I
PARAMETERS OF FEEDING CAVITY

Parameters	Value(mm)	Parameters	Value(mm)
R_g	30	R_m	21
H_g	16.8	H_m	8.2
R_i	1.27	R_o	2.92
W_r	4.6	G	2.2
W_b	9	T_b	1.8
H_d	1.53	H_f	2
L_{g1}	1.35	L_{g2}	2.3

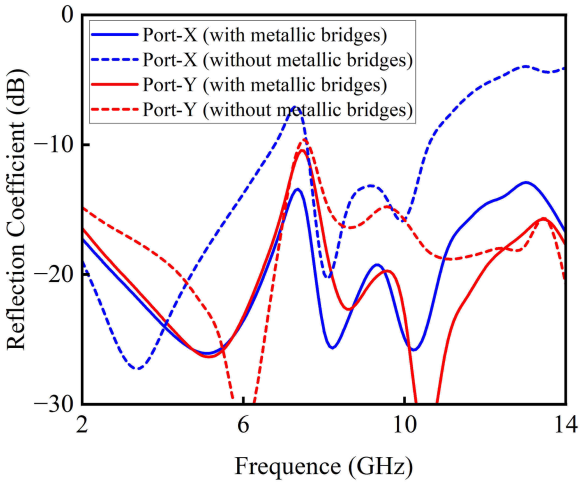


Fig. 2. Effect of metallic bridges on reflection coefficient.

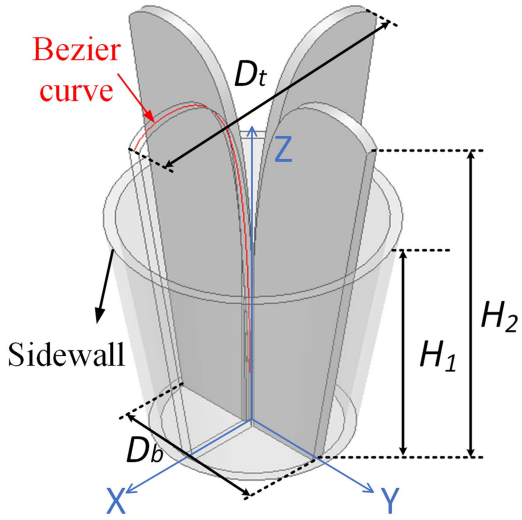


Fig. 3. Structure of flared section.

conventional rectangular block. The introduction of the ring-shaped block enables electromagnetic resonance at the bottom of the cavity and facilitates broadband impedance matching. Since the impedance characteristic is sensitive to the distance between the feeding ports and the matching block, the two feeding ports are positioned with a nonuniform height profile to allow flexible impedance matching. Specifically, additional metallic bridges with height T_m, shown in orange in Fig. 1, are inserted between the ridge of the x-axis feeding port and the matching block within the feeding cavity. These modifications ensure that both feeding ports maintain an optimal distance from the matching block and achieve efficient impedance matching simultaneously.

To verify the effect of the metallic bridges on the reflection coefficient of the feeding cavity, simulations were conducted for two cases: with and without metallic bridges. The simulated results are presented in Fig. 2. The models of the proposed antenna are calculated and analyzed using Ansys HFSS. Since the metallic bridges are placed at the base of the ridge along the x-axis, they exert minimal influence on the reflection coefficient of Port-Y while significantly influencing that of Port-X. Without the metallic bridges, the impedance matching of the feeding cavity is inadequate. After the metallic bridges are introduced, the reflection coefficient within the operating band is reduced below -10 dB, indicating that electromagnetic resonance is established at the bottom of the cavity. The simulation results demonstrate that incorporating the metallic bridges effectively mitigates the impedance mismatch caused by the height disparity between the two feeding ports in the dual-polarized horn antenna. The final parameters of the feeding cavity are listed in Table I.

B. Flared Section

The structure of the horn flare section is shown in Fig. 3. The key parameters of the flared section include the bottom aperture diameter (D_b), the top aperture diameter (D_t), the sidewall height (H_1), ridge height (H_2), and the ridge profile. The aperture and height of the horn flared section directly determine the flare angle. A larger flare angle reduces the low-frequency cutoff, but it also introduces significant phase errors across the aperture, thereby degrading the radiation pattern. In contrast, a smaller flare angle improves the radiation pattern, but results in reduced gain and an increased low-frequency cutoff, which makes it difficult to satisfy the 2 GHz lower operating limit. Consequently, the design of the horn's flared section should be carefully optimized to balance cutoff frequency, radiation pattern stability, and gain, thereby ensuring optimal antenna performance.

As shown by the red line in Fig. 3, to enable precise control of the ridge profile, this work employs Bezier curves to define the ridge geometry, replacing the exponential curves adopted in previous works, which pose inherent challenges in precise control [12], [17], [18]. The start and end points of a Bezier curve are fixed at the first and last control points, while the intermediate control points attract the curve toward them, making Bezier curves well-suited for modeling complex shapes. Furthermore, increasing the curve order introduces additional control points, thereby allowing more accurate curve shaping. Fig. 4 illustrates the final ridge profile, which is defined using an optimized

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on September 29,2026 at 10:06:19 UTC from IEEE Xplore. Restrictions apply.

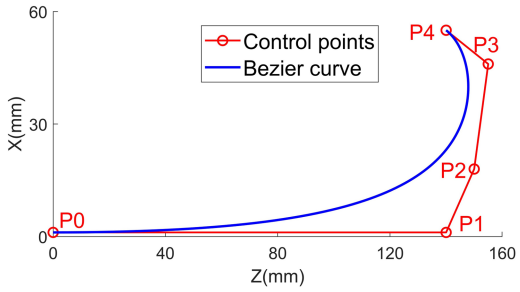


Fig. 4. Optimized Bezier curve of the ridge.

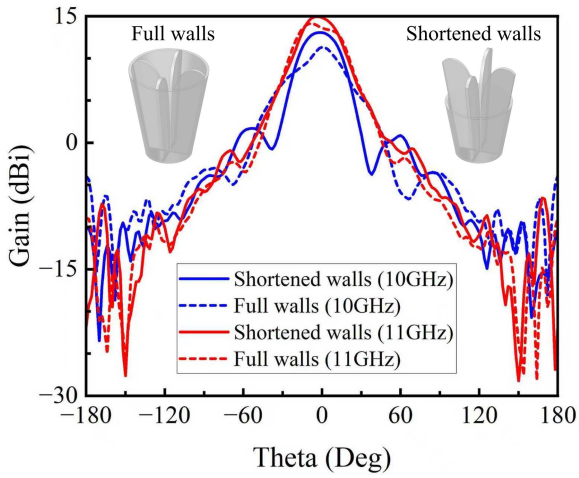


Fig. 5. Effect of sidewall height on gain and radiation pattern.

fourth-order Bezier curve in this work. By optimizing the positions of these control points, a smooth, arc-like transition is achieved, resulting in a more progressive impedance transition. The coordinates (Z, X) of the optimized control points P0 through P4 are (0 mm, 1.1 mm), (140 mm, 1.1 mm), (150 mm, 18 mm), (155 mm, 46 mm), and (140 mm, 55 mm), respectively.

An additional structural enhancement involves the reduction of the sidewall height. This modification was made based on simulations indicating that the E-plane radiation pattern of the full-sidewall antenna exhibited significant distortion at higher frequencies as a result of higher-order mode excitation. In contrast, the shortened sidewalls, which lack metallic barriers, suppressed the generation of higher-order modes, thereby reducing radiation pattern degradation and asymmetry. Fig. 5 illustrates the simulated E-plane radiation patterns of both full- and shortened-sidewall antennas at 10 GHz and 11 GHz for Port-X (Port-Y exhibits similar results and is therefore omitted for brevity). The full-sidewall antenna exhibited asymmetric radiation patterns at 10 GHz and a pronounced dip at 11 GHz, attributable to the excitation of higher-order modes caused by the blockage imposed by the sidewalls. By removing portions of the sidewalls, the excitation of these modes was effectively suppressed, thereby ensuring more symmetric and stable radiation patterns. The optimized dimensions of the flared section are listed in Table II.

III. Measurement Results and Discussion

After completing the antenna design and finalizing the parameters, a prototype was fabricated and its performance was characterized through measurements. The fabricated antenna is

TABLE II
Parameters of Flared Section

Parameters	Value(mm)	Parameters	Value(mm)
D_b	60	D_t	110
H_1	88	H_2	140



(a) (b)

Fig. 6. Fabricated antenna prototype. (a) Perspective view. (b) Top view.

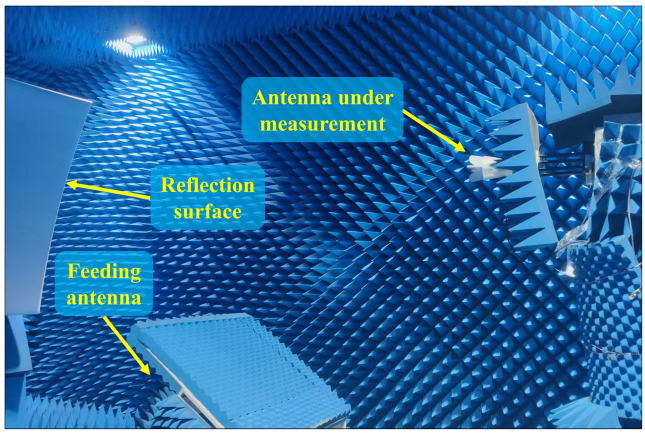


Fig. 7. Measurement environment (compact antenna test range).

shown in Fig. 6, comprising the outer walls, four ridges, and an SMA connector. Each component was manufactured separately and then assembled, with the four ridges secured to the outer walls using screws. The measurement environment is shown in Fig. 7.

Fig. 8 presents the simulated and measured scattering parameters. Good agreement is observed between the simulated and measured results. Across the 2 GHz to 14 GHz band, both $|S_{11}|$ and $|S_{22}|$ remain below -10 dB, indicating satisfactory impedance matching. Moreover, $|S_{21}|$ is consistently below -30 dB, corresponding to port isolation greater than 30 dB. These results demonstrate that the proposed antenna achieves excellent impedance matching and high port isolation. The simulated and measured boresight gain across the entire operating band is presented in Fig. 9. The antenna gain increases progressively with frequency, reaching a maximum of 19.3 dBi. The measured results exhibit strong agreement with the simulations, thereby validating the accuracy of the antenna design.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on September 29,2026 at 10:06:19 UTC from IEEE Xplore. Restrictions apply.

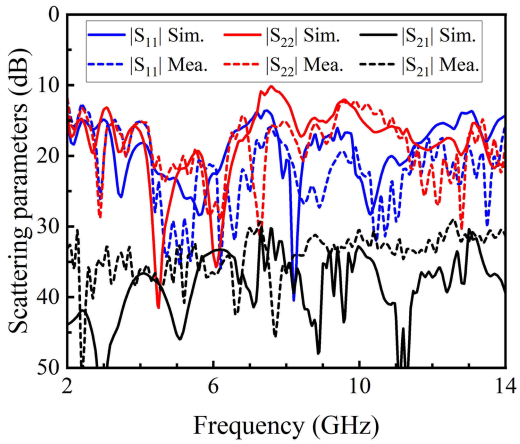


Fig. 8. Simulated and measured scattering parameters of the proposed antenna.

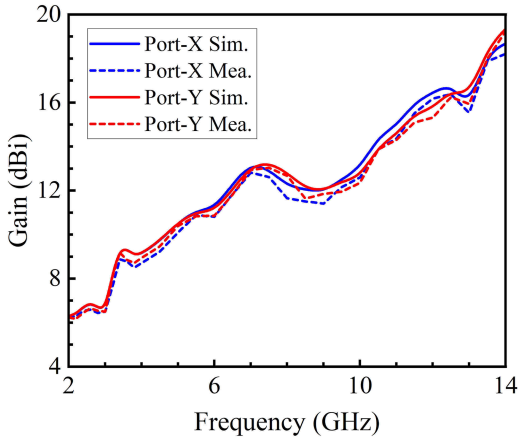


Fig. 9. Simulated and measured gains of the proposed antenna.

The simulated and measured co- and cross-polarized radiation patterns are shown in Fig. 10. Across the entire operating band, the desired stable radiation patterns—free of beam splitting—are successfully observed. This achievement is primarily attributed to the proposed shortened sidewall structure, which effectively suppresses undesired higher-order modes. Besides, low cross-polarization level (XPL) and consistent radiation characteristics of both ports are observed, demonstrating good polarization purity and excellent dual-polarization performance. Furthermore, the close agreement between the measured and simulated results validates the effectiveness and reliability of the antenna design. The slight discrepancy, particularly in the high-frequency range, is primarily attributed to fabrication tolerances and imperfections in the measurement environment.

The comparison between the proposed antenna and other published QRH antennas is presented in Table III. The proposed antenna sustains an exceptionally stable radiation pattern while simultaneously achieving the optimal overall performance across various key performance in terms of wide bandwidth, low VSWR and low XPL. Additionally, reasonable gain and port isolation are also achieved.

IV. Conclusion

A wideband dual-polarized QRH antenna with stable radiation pattern was proposed and investigated in this letter. A metallic ring-shaped matching block replaces the conventional

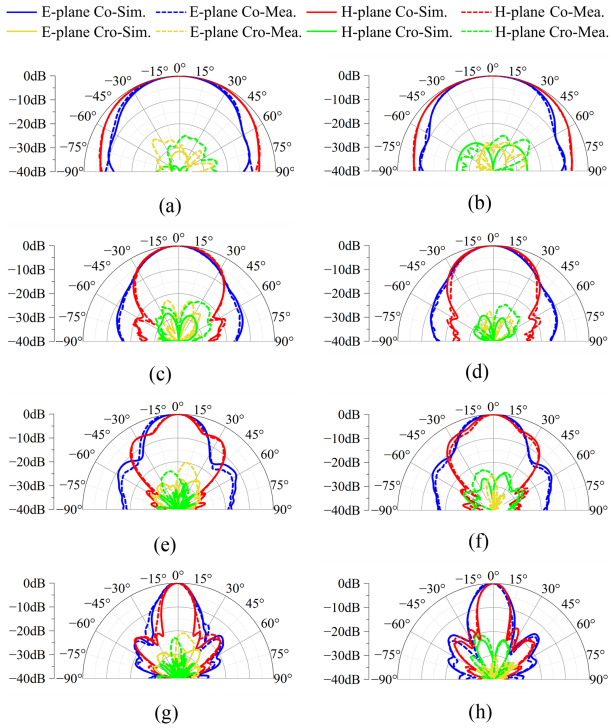


Fig. 10. Simulated and measured normalized radiation pattern of the proposed QRH antenna. (a) Port-X at 2 GHz. (b) Port-Y at 2 GHz. (c) Port-X at 6 GHz. (d) Port-Y at 6 GHz. (e) Port-X at 10 GHz. (f) Port-Y at 10 GHz. (g) Port-X at 14 GHz. (h) Port-Y at 14 GHz.

TABLE III
Comparison of Proposed QRH Antenna With Published Works

Ref.	Band-width (GHz)	VS WR*	Gain* (dBi)	XPL* (dB)	Port Isolation (dB)	Split Radiation Pattern
[19]	1.5-15.5 (10.3:1)	2.6	12	-2	> 30	Yes
[13]	2.4-24 (10:1)	2.1	21.1	N/A	> 25	Yes
[20]	2-6 (3:1)	1.9	N/A	-23	> 30	No
[21]	1-20 (20:1)	2.4	N/A	-5	> 30	Yes
[22]	2.2-14 (6.4:1)	1.9	N/A	-5.2	> 30	Yes
This work	**2-14 (7:1)**	**1.9**	**19.3**	**-20**	**> 30**	**No**

*Maximum value.

structure to ensure wideband impedance match-ing, while two metallic bridges at the ridge base address mismatch from unequal port feeding heights. Furthermore, the ridge profiles and sidewall structure are carefully optimized to ensure smooth impedance transitions and to mitigate radiation pattern distortion by suppressing higher-order modes, respectively. A prototype of the proposed antenna was fabricated and measured, showing good agreement with simulations and validating the design methodology. The antenna achieves superior overall performance, including a 7:1 bandwidth ratio, stable radiation patterns without beam splitting, high gain, low XPL, and high port isolation. These advantages make the antenna a strong candidate for wideband applications in measurements, communications, and sensing.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on September 29,2026 at 10:06:19 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] M. Vidyalakshmi and A. Saha, "Half-TEM horn antenna for EMI/C applications," in *Proc. IEEE Int. WIE Conf. Electr. Comput. Eng.*, Bangalore, India, 2019, pp. 1–4.

[2] H. Hirsch and R. Heinrich, "Simulation and experimental investigations of a TEM horn antenna for RF radiated immunity testing in close proximity," in *Proc. Int. Symp. Electromagn. Compat.*, Angers, France, 2017, pp. 1–5.

[3] L. Song and Y. Nie, "An ultra-wide band exponential flared TEM horn antenna for impulse radar systems," in *Proc. IET Int. Radar Conf.*, Hangzhou, China, 2015, pp. 1–4.

[4] J. Shao, G. Fang, J. Fan, Y. Ji, and H. Yin, "TEM horn antenna loaded with absorbing material for GPR applications," *IEEE Antennas Wireless Propag. Lett.*, vol. 13, pp. 523–527, 2014.

[5] A. S. Turk, A. K. Keskin, M. D. Senturk, A. Magat, M. B. Ozakin, and S. Aksoy, "Ultra wide band TEM horn and reflector antenna designs for down and forward looking ground penetrating radars," in *Proc. 7th Int. Workshop Adv. Ground Penetrating Radar*, Nantes, France, 2013, pp. 1–6.

[6] C. A. Balanis, *Modern Antenna Handbook*. Hoboken, NJ, USA: Wiley, 2011.

[7] "Very-High-Frequency Techniques II," Radio Research Laboratory, Harvard Univ., Cambridge, MA, USA, 1947, pp. 725–728.

[8] J. L. Kerr, "A very broad band low silhouette antenna," U.S. Army Electron. Command (ECOM), Fort Monmouth, NJ, USA, Tech. Rep. ECOM-3087, 1967.

[9] J. Kerr, "Short axial length broad-band horns," *IEEE Trans. Antennas Propag.*, vol. 21, no. 5, pp. 710–714, Sep. 1973.

[10] B. Jacobs, J. W. Odendaal, and J. Joubert, "An improved design for a 1–18 GHz double-ridged guide horn antenna," *IEEE Trans. Antennas Propag.*, vol. 60, no. 9, pp. 4110–4118, Sep. 2012.

[11] Z. Yue, Y. Liu, and S. Gong, "A novel 2–18 GHz double-ridged horn antenna with simple structure," *Microw. Opt. Technol. Lett.*, vol. 60, no. 7, pp. 1787–1794, 2018.

[12] C. Wang, E. Li, Y. Zhang, and G. Guo, "Ridged horn antenna with adjustable metallic grid sidewalls and cross-shaped back cavity," *IEEE Antennas Wireless Propag. Lett.*, vol. 15, pp. 1221–1225, 2016.

[13] Y. Ma et al., "A 10:1 bandwidth cryogenic quadruple-ridged flared horn design for reflector antennas in radio astronomy," *IEEE Access*, vol. 8, pp. 81101–81115, 2020.

[14] X. Zhang, Z. Wang, C. Pan, X. Ba, and Y. Wang, "Dual-polarized inverted quad-ridged flared horn antenna with one-decade bandwidth for OTA testing," *IEEE Antennas Wireless Propag. Lett.*, vol. 21, no. 6, pp. 1233–1237, Jun. 2022.

[15] C. Bruns, P. Leuchtmann, and R. Vahldieck, "Full wave analysis and experimental verification of a broadband ridged horn antenna system with parabolic reflector," in *Proc. IEEE Antennas Propag. Soc. Int. Symp.*, Boston, MA, USA, vol. 4, 2001, pp. 230–233.

[16] C. Bruns and P. Leuchtmann, "Analysis and simulation of a 1–18 GHz broadband double-ridged horn antenna," *IEEE Trans. Electromagn. Compat.*, vol. 45, no. 1, pp. 55–60, Feb. 2003.

[17] M. Abbas-Azimi, F. Arazm, J. Rashed-Mohassel, and R. Faraji-Dana, "Design and optimization of a new 1–18 GHz double ridged guide horn antenna," *J. Electromagn. Waves Appl.*, vol. 21, no. 4, pp. 501–516, 2007.

[18] M. Gerber, J. W. Odendaal, and J. Joubert, "DRGH antenna with improved gain and beamwidth performance," *IEEE Trans. Antennas Propag.*, vol. 68, no. 5, pp. 4060–4065, May 2020.

[19] J. Flygare and M. Pantaleev, "Dielectrically loaded quad-ridge flared horn for beamwidth control over decade bandwidth—Optimization, manufacture, and measurement," *IEEE Trans. Antennas Propag.*, vol. 68, no. 1, pp. 207–216, Jan. 2020.

[20] Z. Zhao, Z. Wang, J. R. Kelly, and S. Gao, "Balanced feed low cross-polarization quad-ridged horn antenna with center cross fusiform loaded," *IEEE Trans. Antennas Propag.*, vol. 69, no. 9, pp. 6004–6009, Sep. 2021.

[21] J. Flygare et al., "Beyond-decade ultrawideband quad-ridge flared horn with dielectric load from 1 to 20 GHz," *IEEE Trans. Antennas Propag.*, vol. 71, no. 3, pp. 2110–2125, Mar. 2023.

[22] Ó. García-Pérez et al., "A modular approach for the design of quadruple ridged flared horn antenna feeds," *IEEE Access*, vol. 12, pp. 69642–69651, 2024.

Authorized licensed use limited to: XIDIAN UNIVERSITY. Downloaded on September 29,2026 at 10:06:19 UTC from IEEE Xplore. Restrictions apply.